Experiment-18 : OPTICAL SPECTRUM AND OSNR ANALYSIS USIN OPTISYSTEM

AIM:

To design and simulate an optical communication system using OptiSystem and analyse its optical spectrum and OSNR performance.

OBJECTIVES:

1. Design an optical communication system using OptiSystem.
2. Study how source power affects optical spectral intensity.
3. Analyze how fiber length affects attenuation, dispersion, and nonlinear effects.
4. Study the combined effect of source power and fiber length.

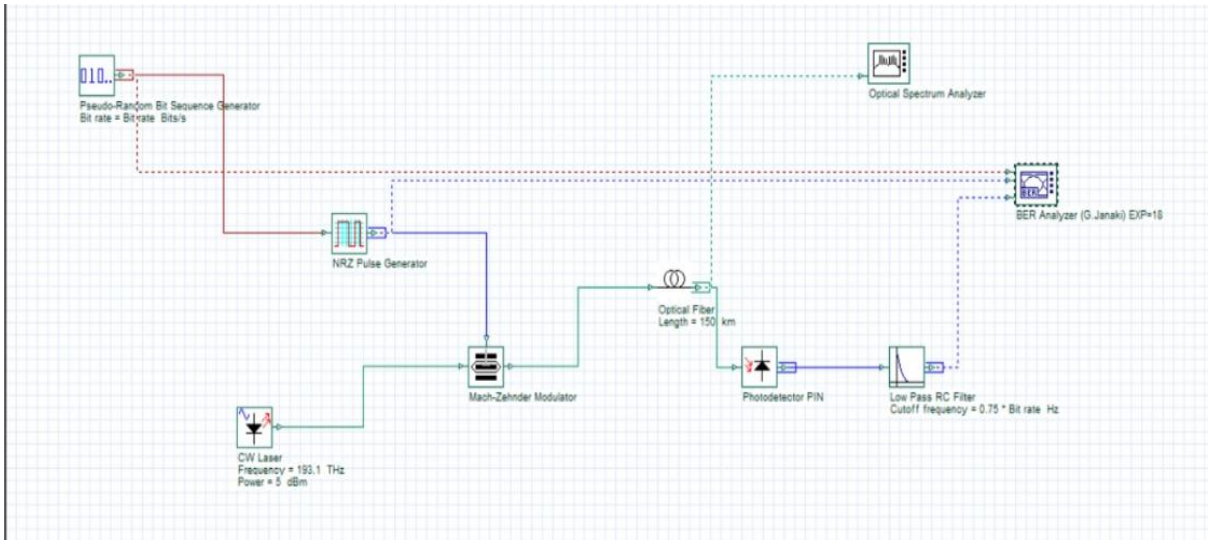
SOFTWARE REQUIRED:

- Optisystem

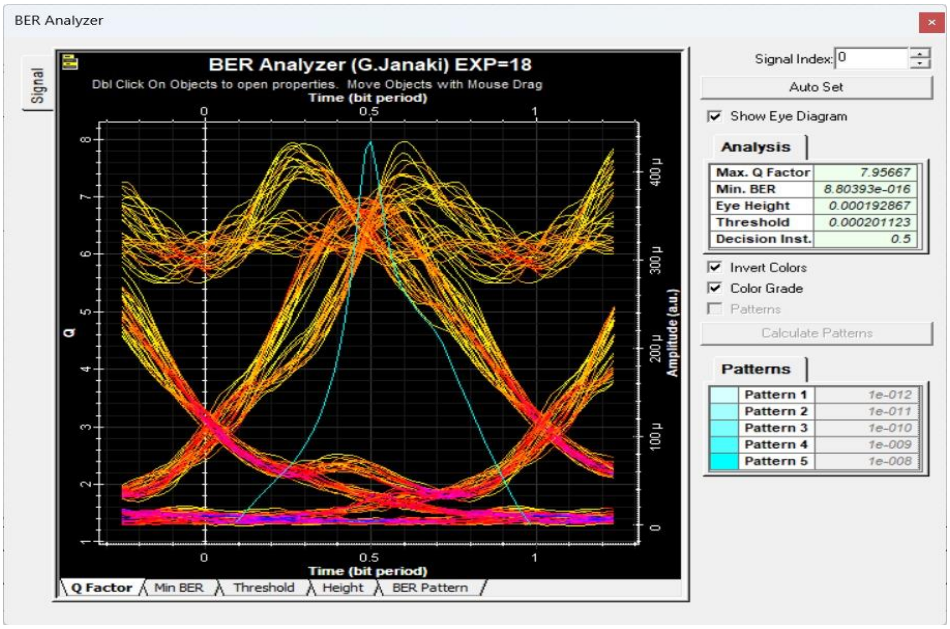
EXPERIMENT PROCEDURE:

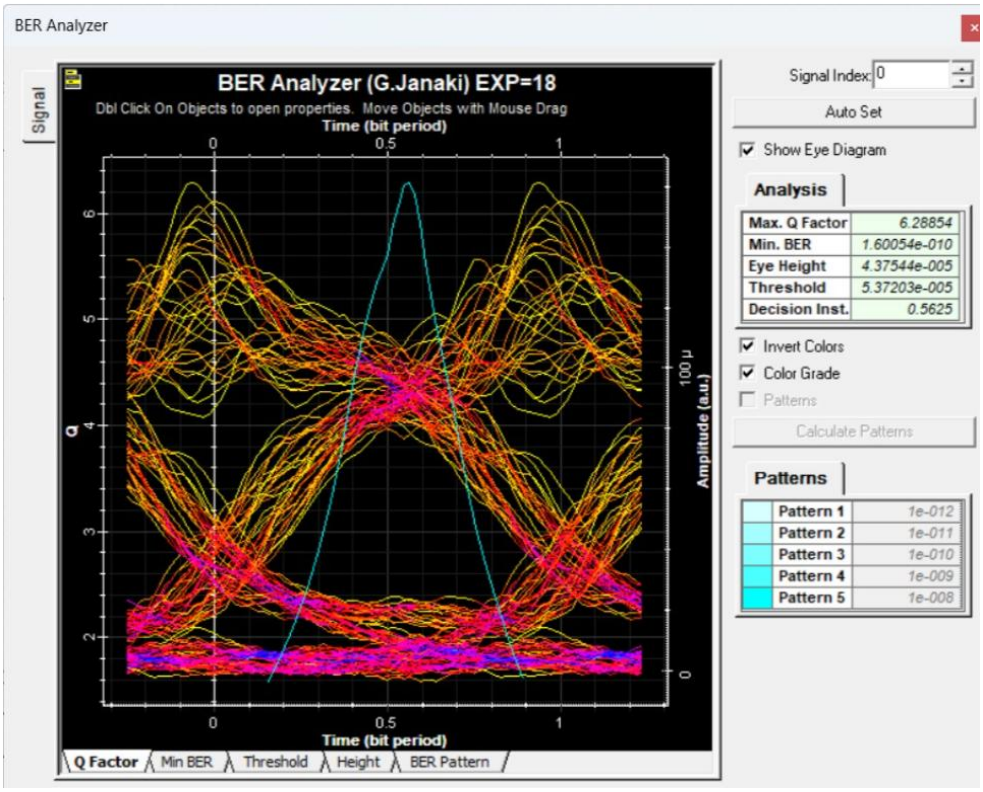
1. Design the optical link in OptiSystem using laser, NRZ generator, MZ modulator, fiber and PIN photodiode
2. Connect the Optical Spectrum Analyzer to observe the optical spectrum.
3. Set wavelength to 1550 nm
4. Vary source power and fiber length according to the test cases
5. Run the simulation and record spectral power, spectral width and received power
6. Compare the results for different power and fiber-length values

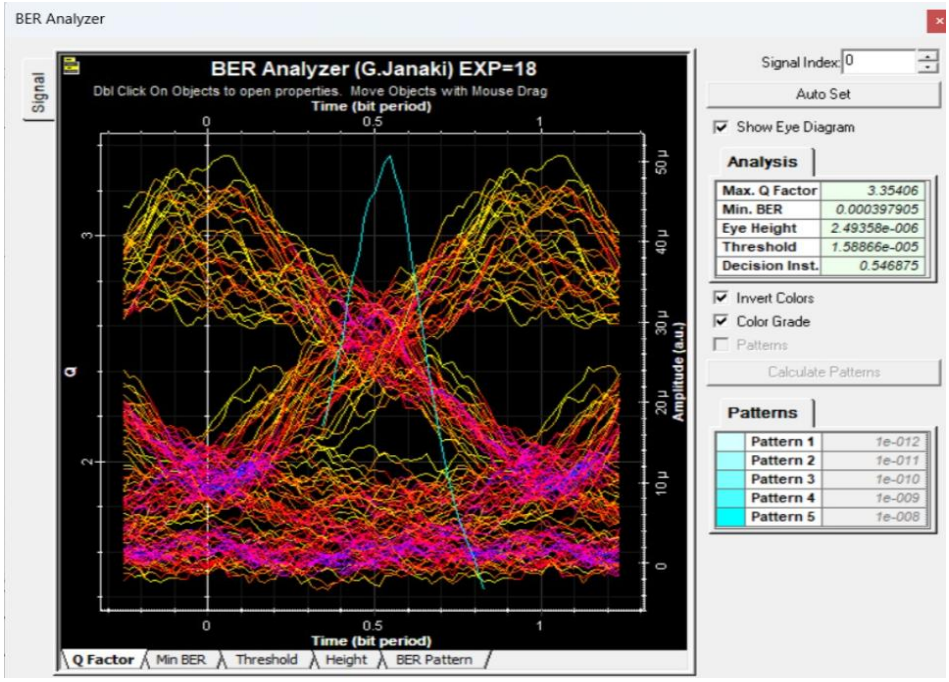
BLOCK DIAGRAM:

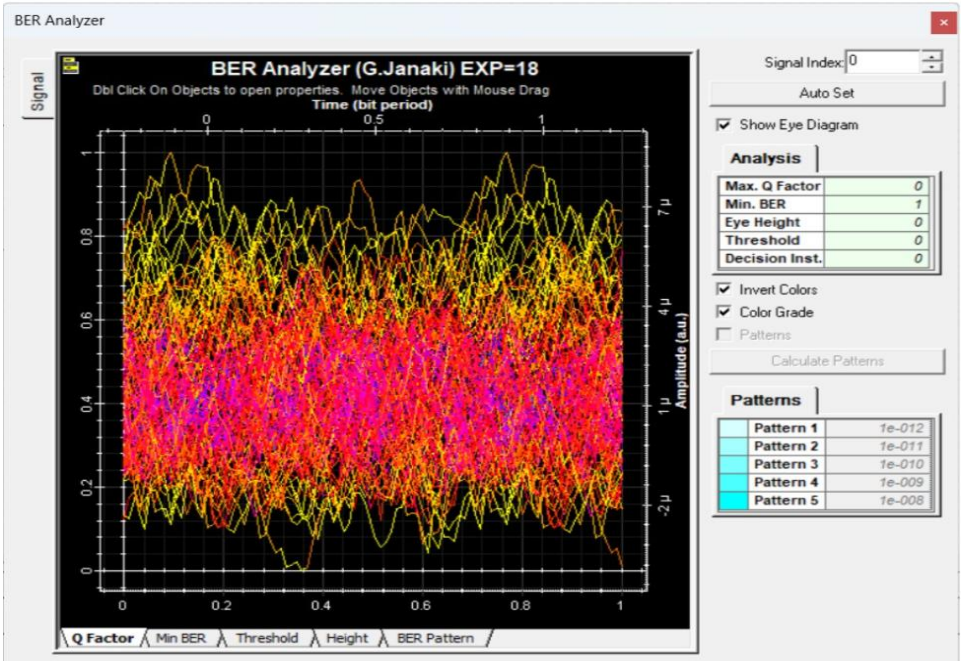


OUTPUT WAVEFORMS:









REFLECTION:

This project helped to understand optical spectrum analysis, OSNR measurement and the effect of noise on optical signal quality. It improved practical knowledge of optical communication system simulation using OptiSystem.

RUBRICS:

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
System Design	Correct optical system design.	Minor errors.	Partial setup.	Incorrect setup.	20%
Parameter Configuration	All parameters correctly set.	Few errors.	Some values missing.	Incorrect values.	20%
Measurement & Data Collection	Accurate OSNR and spectrum readings.	Minor missing data.	Partial readings.	Incorrect readings.	20%
Data Analysis	Detailed spectrum and OSNR analysis.	Basic analysis.	Limited analysis.	No proper analysis.	20%
Report & Conclusion	Complete report with graphs and results.	Good report.	Incomplete report.	Poor report.	20%

RESULT:

The optical communication system was successfully simulated, and the optical spectrum and OSNR values were analysed.